

## PROCEED v6.2

Phenotypic Rates of Change Evolutionary and Ecological Database

RatesDB – Metadata

## General

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Ecology and Evolution

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### Data Contributions

Carnegie Museum of Natural History, Maine Department of Marine Resources, NS-ITBS surveys ([www.ices.dk/marine-data/dataportals/Pages/DATRAS.aspx](http://www.ices.dk/marine-data/dataportals/Pages/DATRAS.aspx)), Powdermill Avian Research Station

Technical

Language: English

Decimal point= “.”

Not available data (NA)= “” (i.e. empty cell)

Description, Inclusion/Exclusion criteria

Phenotypic Rates of Change Evolutionary and Ecological Database (PROCEED) is an ongoing compilation of rates of phenotypic change. We compile and add data regularly to the data set. This dataset is continually being updated as more people ask it include new variables.

This database includes studies that measure intraspecific change in quantitative (continuous or counting) traits and report the time elapsed from the onset of environmental novelty or refer to a historical or biological event reported in other sources (e.g., a mine opening, a well documented biological invasion). The maximum elapsed time between the environmental change and the sampling was no longer than 500 years. However, most of the observations correspond to less than 100 years of elapsed time (3rd quartile 89.5 years) and only a few cases correspond to more than 200 years of elapsed time. The time measured as years and generations were positively correlated, but for a given time in years, there is a wide range of variability in generations, approximately two orders of magnitude. The included studies followed a single population through time (allochronic design) or compare two (or more) populations that diverged at a known time (synchronic design). The studies may record the differences among populations in the field (here called phenotypic studies) or in common growing conditions (here called genetic studies). The database records the change in diverse trait types and environmental changes.

The database includes systems in natural or experimental conditions but in which reproduction was not manipulated (i.e. NO artificial selection is included). The experimental systems manipulate the environment of the populations being studied but do not select traits deliberately (e.g. an herbivory exclusion experiment, where the type and load of herbivory are manipulated and the populations evolve under this new condition). The database does not include systems where the phenotypic change is due to interspecific hybridization, polyploidy, or other chromosomal mutations.

PROCEED provides information to compute two popular and complementary evolutionary rates, Darwins and Haldanes, in contemporary timescales. PROCEED also provides a set of biological and methodological variables aimed at modeling and understanding the variability in those rates of phenotypic change.

The Darwins ( $D$ ) rate of change is the proportional change in a quantitative trait ( $x$ ) over time (Haldane 1949).

$$D = \frac{1}{x} \cdot \frac{dx}{dt}$$

The integral of this equation across a discrete period ( $t=t_2-t_1$ ) is the difference between the natural logarithm of a trait value at a particular time (or in a particular population) and the natural logarithm of the same trait at some time in the past (or in another population), over the length of time (typically expressed as millions of years).

$$D = \frac{\left[ \ln(\bar{x}_2) - \ln(\bar{x}_1) \right]}{\left[ t_2 - t_1 \right]}$$

$$D = \frac{\ln(\bar{x}_2/\bar{x}_1)}{t}$$

This means that the value of the trait at time 2 is a function of the value at time 1 as follows:

$$\bar{x}_2 = \bar{x}_1 \cdot e^{D \cdot t}$$

The Haldanes (H) rate of change (Haldane 1949) scales the magnitude of change by the amount of variation in the trait. Lerman (1965) further articulated this measure by expressing rates as “differences between the population means in units of standard deviation.” Gingerich (1993) formulated a similar metric and dubbed it the Haldane (the same measure is designated D' H by Lynch 1990). A simple formulation of the Haldane is:

$$H = \frac{\frac{(\bar{x}_2 - \bar{x}_1)}{SD_x}}{g}$$

where  $\bar{x}_2$  and  $\bar{x}_1$ , represent mean trait values for each of two populations (synchronic) or for a single population at two different times (allochronic),  $SD_x$  is the pooled standard deviation, and  $g$  is the number of generations separating the populations or samples (years divided by generation time). For many traits, especially morphological characters, standard deviations are expected to increase with the mean (i.e., coefficient of variation remains relatively constant). For such traits, raw data should be transformed to natural logarithms ( $\ln$ ), which will reduce heteroscedasticity (Wright 1968). However, estimated rates of change calculated using  $\ln$  means and standard deviations do not differ appreciably from those calculated using non-transformed means and standard deviations (Hendry & Kinnison 1999).

Darwins and Haldanes differ in two fundamental ways. First, Darwins specify the rate of proportional change in units of the initial value assuming an exponential rate of change, whereas Haldanes specify the rate of change in standard deviation units. Second, the time interval is measured in

million of years for Darwins and in generations for Haldanes. In principle (but not in practice), these two differences are independent of each other. Indeed, a proportional change could just as easily be specified per generation or a standardized change per year. If an investigator's goal is to measure a change in some organism that is relevant to time dependent human interests, then proportionate change and a time unit based on years may be the most desirable characteristic of a rate measure (i.e., Darwins). However, if the goal is to understand how a population responds to environmental change or to estimate the intensity of selection, then standardizing by the trait's variation and using a time scale more relevant to the organism (i.e., generations) will provide a more appropriate rate measure (i.e., Haldanes).

In accounting for variation, Haldanes have a better grounding in the evolutionary process, but this property adds additional complexity when comparing genetic and phenotypic studies. Environmentally-induced variation may be greater in the wild than it is under common conditions. In such cases, genetic Haldanes will be higher than phenotypic Haldanes for the same absolute magnitude of change. In contrast, if individuals with extreme phenotypes are selected against in the wild, but not in the laboratory, genetic Haldanes may be lower than phenotypic Haldanes for the same magnitude of change.

The accuracy of an estimated evolutionary rate depends on the accuracy of estimated elapsed time (years for Darwins, generations for Haldanes). Estimating the number of years that have passed might seem easy, but it is not without ambiguity in many instances. Small errors in the estimated year of population founding, for example, can translate into measurable differences in evolutionary rates when the total time interval is relatively short. Obtaining good estimates of the time interval in generations is often more difficult because of the added uncertainty in the estimation of generation time. As a result, rates estimated in Haldanes may be less accurate than those estimated in Darwins, regardless of how well each measure reflects the evolutionary process.

The dimension in which a trait is measured (e.g., length, area, volume) influences rates of evolution calculated using Darwins (Gingerich 1993). This dimension dependence arises because as length increases, area increases as the square function of that change, and volume as the cube function of that change (assuming a cubic trait). On the other hand, Haldanes are not dimension dependent. Phenotypic data is commonly measured on ratio or interval scales. Data on a ratio scale have a constant interval between adjacent units, and the measurement scale has a precise zero point corresponding to a null quantity. Ratio scale data have the property that doubling a value doubles the actual quantity. In contrast, data on an interval scale have a constant interval between adjacent units, but the zero point is arbitrary (e.g., time of day). Darwins are not appropriate for specifying rates of evolution for interval scale traits. Behavioral and phenological traits are often measured on an interval scale, such as migratory timing, or flowering time. Darwins cannot be used to estimate rates in those cases because, for example, in the migration case, What is day zero? If it is assumed to be the first day of the year, the rate of change will be different from the rate of change for the same amount of change if day zero is assumed to be June 1. On the contrary, Haldanes is still suitable for calculating evolutionary rates on interval scales.

In a similar way to the issues with the interval scales, for morphological traits, the choice of the landmarks may impact the Darwins if an evolutionary change is not proportional along its entire structure being measured (i.e., the change occurs in a smaller section of the structure). In those cases, the landmark operates as a misleading zero. This discrepancy arises because the same change in only part of the structure will appear smaller when measured from a more distant zero point. Haldanes are not as strongly influenced as Darwins by the choice of different landmark points when part of the trait is invariant. Furthermore, both Darwins and Haldanes are ill-equipped to handle rates of change in nonmetric characters, such as color morphs, presence or absence of a character state, behavioral options, or alternative life histories.

Darwins and Haldanes provide different but complementary measures of evolutionary rate. Because both specify a phenotypic difference per unit time, rates calculated using each will be correlated in general. However, some studies might have relatively low Haldanes rates but high Darwins rates, or the opposite, because of differences in the time units (years vs generations), or because of the unit of change (trait initial value vs trait variance). Due to the different conceptual basis of the two measures and to the different information conveyed by each, it is worth presenting both when possible. It is also important to acknowledge the limitations of each measure as discussed above.

According to previous definitions and considerations, Darwins cannot be computed if the trait is not measured in a ratio scale, or if the average mean of sample 1 (in a raw scale) is equal to zero, or if it is a missing value, or if the ratio between mean 2 and mean 1 (in a raw scale) is a negative value. Additionally, we did not compute Darwins if the trait measurement scale is unknown whether it is in the raw scale or any transformation (e.g. logarithmic) was applied. Only data in the raw or ln-transformed scale were included. Here, Darwins numerators were computed in the raw scale of the trait. If the scale of the trait was ln-transformed (ln: natural logarithm), the Darwins numerator was computed according to the Lynch (1990) approximation. That is, the natural logarithm of the average of the trait at a given time is approximated as the average of the natural logarithm of the trait plus the half of the variance of the natural logarithm of the trait:

$$\ln(\bar{x}_t) = \ln(\bar{x}_t) + 0.5 \cdot \sigma_{\ln(x_t)}^2$$

As a consequence, the records expressed in ln-transformed scale also requires the standard deviation to be computed.

According to previous definitions and considerations, Haldanes cannot be computed if the standard deviation of sample 1 or 2, or any of the sample sizes are lacking. Also, it cannot be computed if the generation time is unknown.

The datasets PROCEEDv6.2\_species\_genome, PROCEEDv6.2\_taxonomy, and PROCEEDv6.2\_species\_GT are complementary information with additional information about the genome size, the taxonomy, and generation time respectively of the species in the Rates and TimeSeries datasets. Finally, PROCEEDv6.2\_BasicCode compiles R code for some basic handling

of PROCEEDv6.2 and examples to perform some basic analyses and graphics. See the metadata of each dataset for further details.

Variables:**es\_ID**

Unique identifier for each individual rate.

**ref\_ID**

Unique identifier for each primary source of information (paper, thesis, report, etc).

**sys\_ID**

Unique identifier for each study system (see “system” for the definition).

**released\_sys**

sys\_ID of the system originating the new system after removal/extinguishment/stop of the driver/disturbance. For example, the system s445 (*Ovis canadensis* in Ram Mountain, Alberta, Canada when a hunting moratorium was placed starting in 1996) is the same population of system s122 (*Ovis canadensis* in Ram Mountain, Alberta, Canada when hunting was allowed up to 1995). So, for the system s445 “released\_sys”=“s122”. On the contrary, for the system s122 “released\_sys”=“NA” (because s122 was not derived from any other system in the database). Future studies with similar situations (i.e. a single population in periods with different evolutionary drivers) should include, in “released\_sys” of the new condition, the sys\_ID of the former condition.

**released\_es**

es\_ID of the case originating the new case after removal/extinguishment/stop of the driver/disturbance in the system. A system being released from the original driver generates a new system. If more than one trait is measured in both systems (the original and the new one), “released\_es” links the cases in the new system with the measurements of the same trait in the original systems. For example, in the systems s122 and s445, two traits were measured: mean weight (kg) and mean horn length (cm). In s122, the change in mean weight is the case “es\_ID”=“es2798” and the change in mean horn length is the case “es\_ID”=“es2799”. So, for the case es9277 (change in mean weight in the system s445), “released\_es”=“es2798”. For the case es9278 (change in mean horn length in the system s445), “released\_es”=“es2799”. On the contrary, for es2798 and es2799 (system s122), “released\_es”=“NA” (because s122 was not derived from any other system in the database). Future studies with similar situations (i.e. a single population in periods with different evolutionary drivers) should include, in “released\_es” of the new condition, the es\_ID of the same traits in the former condition.

**sp\_pub**

Species name (or subspecific taxon name) of the population(s) being assessed in each case, as was published in the primary source of information.

#### **sp\_ncbi**

Species or subspecific taxon name of the population(s) being assessed in each case, according to the National Center for Biotechnology Information (<https://www.ncbi.nlm.nih.gov/Taxonomy/Browser/wwwtax.cgi?mode=Root>) in August 2025. Two species were not found in the NCBI database, *Stipa occidentalis* and *Diaptomus sanguineus*, these names were checked in The Plant List (<http://www.theplantlist.org/>) and WORMS (<https://www.marinespecies.org/index.php>) respectively. The supplementary table “PROCEEDv6.2\_taxonomy.csv” contains the complete list of “sp\_pub” and its “sp\_ncbi”.

#### **taxa**

This is a coarse classification, not necessarily a clade (monophyletic group). Current categories are: **Annelid, Mollusc, Arthropod, Fish, Amphibian, Mammal, Bird, Reptile, Plant.**

#### **phylum**

Taxonomic clade of rank “phylum” including the studied species, according to the National Center for Biotechnology Information (<https://www.ncbi.nlm.nih.gov/Taxonomy/Browser/wwwtax.cgi?mode=Root>) last retrieved August 2025. The supplementary table “PROCEEDv6.2\_taxonomy.csv” contains the complete list of “sp\_ncbi” and its “phylum”.

#### **class**

Taxonomic clade of rank “class” including the studied species, according to the National Center for Biotechnology Information (<https://www.ncbi.nlm.nih.gov/Taxonomy/Browser/wwwtax.cgi?mode=Root>) last retrieved August 2025. The supplementary table “PROCEEDv6.2\_taxonomy.csv” contains the complete list of “sp\_ncbi” and its “class”.

#### **order**

Taxonomic clade of rank “order” including the studied species, according to the National Center for Biotechnology Information (<https://www.ncbi.nlm.nih.gov/Taxonomy/Browser/wwwtax.cgi?mode=Root>) last retrieved August 2025. The supplementary table “PROCEEDv6.2\_taxonomy.csv” contains the complete list of “sp\_ncbi” and its “order”.

#### **system**

It is a population or a set of populations, of a given species, experiencing a specific driver/disturbance. If the design is synchronic, the system includes both diverging

populations (or sets of populations). Each system has its own unique combination of species, disturbance/driver and location. Within a given system, you can have multiple traits – for example, tarsus length and fledging date. We suggest keeping it general. Please note you can have both allochronic and synchronic data within the same system (e.g. Leaver 2012, Biol. J. Linnean. Soc. 107: 494-509). Example systems: “*Salmo trutta* (brown trout) monitored for 37 y following fish ladder construction (1966) in River Gudbrandsdalslågen, Norway”; “*Troglodytes troglodytes* (Eurasian Wren) migration arrival date monitored in its native range in Europe”; “*Oncorhynchus gorbuscha* (pink salmon) long term trends on northwest coast of US”.

### **experimental**

States if the phenotypic change is driven by experimentally set conditions (**N**=no; **Y**=yes). It is not artificial selection (i.e. selection of specific attributes by choosing breeders and manipulating mating) but a process of natural selection in an experimental environment.

### **exp\_type**

When “experimental”=“N”, “exp\_type”=“NA” (empty cell). When “experimental”=“Y”, the following categories were applied:

- **01**: A new environmental condition is imposed on a natural population, in the field, in situ.
- **03**: A new environmental condition is imposed on a natural population, in the field, ex situ (transplant).
- **05**: A new environmental condition is imposed on a natural population, in a common garden/greenhouse experiment.
- **07**: A new environmental condition is imposed on an artificially generated population in the field. Here, the artificial population is a set of genotypes from the natural population, evenly represented by (full or half) sibling families.
- **09**: A new environmental condition is imposed on an artificially generated population in a common garden experiment. Here, the artificial population is a set of genotypes from the natural population, evenly represented by (full or half) sibling families.
- **11**: A new environmental condition is imposed on an artificial population (it was obtained by randomly crossing F2 individuals from a set of interbreeds between contrasting genotypes selected from a natural population) in a common garden experiment.

### **comments\_on\_system**

Any comments about the system should be stated here.

### **c\_value**

A measurement of genome size, it is the amount (in picograms) of DNA contained within a haploid nucleus (e.g. a gamete) or one half the amount in a diploid somatic cell. Detailed information of the variable, sources, ploidy level, analytical methods, etc in the supplementary table “PROCEEDv6.2\_species\_genome.csv” and “PROCEEDv6.2\_species\_genome\_README.txt”.

#### **sex**

Sex of the individuals measured (male, female, asexual, mixed=male+female, NA(empty cell)=sex not reported in the study).

#### **sample1**

Identification of Sample 1 (e.g. year, population, identifier given by author(s)). In the case of allochronic studies, Sample 1 should always be the earlier sample. In the case of synchronic studies, sample 1 should always be the source population (e.g. in the case of introductions or range expansions, Sample 1 should be the site “before introduction or expansion”) or the population(s) in the original/ancestral environment condition (e.g. in the case of pollution, sample 1 should be the population in the non-polluted area; in the case of response to introduction, sample 1 should be the population in the area where the introduced species is not present). When the traits are measured in a common garden experiment with different conditions, in more than one common garden location, or in reciprocal transplant experiments, these conditions are noted here (e.g. sample1: “Control origin, control treatment”, sample2: “Polluted origin, Control treatment”; sample1: “Control origin, Drought treatment”, sample2: “Polluted origin, Drought treatment”; sample1: “Native populations, Jokers Hill common garden”, sample2: “Introduced populations, Jokers Hill common garden”; sample1: “Native populations, Montpellier common garden”, sample2: “Introduced populations, Montpellier common garden”).

#### **sample2**

Identification of Sample 2 (e.g. year, population, identifier given by author(s)). In the case of allochronic studies, Sample 2 should always be the later sample. In the case of synchronic studies, Sample 2 should always be the later populations (e.g. in the case of introductions or range expansions, Sample 2 should be the site “after introduction or range expansion”) or the population in the new environmental condition (e.g. in the case of pollution, sample 2 should be the population in the polluted area; in the case of response to introduction, sample 2 should be the population in the area where the introduced species is present). When the traits are measured in a common garden experiment with different conditions, or in more than one common garden location, or in reciprocal transplant experiments, these conditions should be log here (e.g. sample1: “Control origin, control treatment”, sample2: “Polluted origin, Control treatment”; sample1: “Control origin, Drought treatment”, sample2: “Polluted origin, Drought treatment”; sample1: “Native populations, Jokers Hill common garden”, sample2: “Introduced populations, Jokers Hill common garden”; sample1: “Native

populations, Montpellier common garden”, sample2: “Introduced populations, Montpellier common garden”).

#### **samp1\_anc**

(yes/no) If sample 1 was from the ancestral population or the population from the ancestral place/source/condition, “samp1\_anc”=“**yes**”. If there is no ancestral-to-derived relationship between samples, then “samp1\_anc”=“**no**”.

#### **sample1\_latitude**

Latitude in decimals (to a minimum of three decimal places if possible) of the location of the “sample1” population. Note that South is negative. If the lat/longs are not given please look up via other studies of the same system, Wikipedia, Google Maps, or OpenStreetMap. You can use these websites to convert from degrees or UTM:

- For URL: [http://andrew.hedges.name/experiments/convert\\_lat\\_long/](http://andrew.hedges.name/experiments/convert_lat_long/)
- For UTM: [http://www.engineeringtoolbox.com/utm-latitude-longitude-d\\_1370.html](http://www.engineeringtoolbox.com/utm-latitude-longitude-d_1370.html)

#### **sample1\_longitude**

Longitude in decimals (to a minimum of three decimal places if possible) of the location of the “sample1” population. Note that West is negative. If the lat/longs are not given please look up via other studies of the same system, Wikipedia, Google Maps, or OpenStreetMap. You can use the above websites to convert from degrees or UTM.

#### **sample2\_latitude**

Same as “sample1.latitude” but for the location of “sample2” populations.

#### **sample2\_longitude**

Same as “sample1.longitude” but for the location of “sample2” populations.

#### **coordinate\_means**

If the coordinates for sample1 and sample2 are the average of coordinates of more than one population (**Y**) or not (**N**).

#### **coordinate.notes**

Source of coordinate information and any other relevant information about the coordinates.

#### **design**

- **Allochronic**: same population/different time points, longitudinal design.
- **Synchronic**: Different populations with known divergence time.

**genphen**

It distinguishes studies measuring individuals that developed in nature (“**Phenotypic**”) from those based on measurements of individuals bred in common-garden conditions or applying other quantitative-genetic methods (“**Genetic**”). Phenotypic does not mean it is not genetic; we just do not have the information and it does not automatically imply plasticity. Quantitative genetic methods (heritability does not count) are usually based on animal model methods and calculate breeding values which is the genetic contribution to a trait (Wilson et al. 2010). Common garden experiments performed from vegetative (clonal) propagules collected in the field (e.g. tiller, an anatomic unit in grasses) are still “Phenotypic”.

**years**

If the study is “design”=“Allochronic”, the number of years from the start time point to the end time point. If the study is “design”=“Synchronic”, the number of years from the divergence point to the date of sampling. If it is less than one year, a decimal is used (number days of study/365).

**gtime**

Generation time expressed in years. We define the generation time as the time (in years) in which a population replaces itself. The supplementary table “PROCEEDv6.2\_species\_GT.csv” summarizes all the information about generation time for each species, compiled in the current version of PROCEED. The file “PROCEEDv6.2\_species\_GT\_README.pdf” provides definitions, criteria, and procedures applied in different situations.

**gtime\_notes**

Source of the “gtime” and any other comment related to “gtime” and “generations”.

**generations**

The number of generations that have elapsed given the number of years that have elapsed for a given system. It can be calculated as: years/gtime.

**trait\_description**

Trait being quantified in the words of the authors of the paper. If the name of the trait is not self descriptive, please include the definition from the paper (e.g. shell slenderness ( $\log_{10}[\text{width}] / \log_{10}[\text{length}]$ )). If units are given, please include. Each trait has its own row.

**trait\_type**

Types of traits based on Kingsolver and Diamond (2011) but with some additional categories. Current categories are:

- **phenology**: the timing of cyclic and seasonal life-history events (e.g. flowering date, laying date, and hatching date). Do not confound with “otherLH”. For example, the time spent in a given stage of the life cycle (e.g. larval stage) or the time to reach a given stage of the life cycle (e.g. time to maturity, time to first flower) are life-history traits, not phenology. “data\_type” for “phenology” traits is typically a date while for “otherLH” it is time.

- **behaviour**: “Broadly speaking, animal behavior includes all the ways animals interact with other members of their species, with organisms of other species, and with their environment. Behavior can also be defined more narrowly as a change in the activity of an organism in response to a stimulus, an external or internal cue or combination of cues.”  
(<https://www.khanacademy.org/science/ap-biology/ecology-ap/responses-to-the-environment/a/intro-to-animal-behavior>)

- **physio**: traits related to physiology or a biochemical parameter (e.g. enzymatic activity). Example physiology traits include metabolism, bite force, leaf toughness, decomposition rate, stoichiometry (e.g. C:N), concentration or amount (yield) of a given chemical compound in the body (or part of the body), diet (trophic level, diet, food type), severity or level of affectation by a disease/pathogen/parasite/herbivore (e.g. Fungal infection % of plants, % leaf consumed), dose/concentration/time exposed to a given substance/condition (e.g. pesticide, drought, light, temperature, etc) needed to produce some effect (e.g. death, flowering, etc.), performance of a standard parasite/herbivore/competitor feeding/competing on/with the focal population (e.g. Pupal weight of the herbivorous *Trichoplusia ni* feeding on native vs introduced populations of *Eschscholzia californica*; or Leaf feeder (*Galerucella pusilla*, L1-Imago) survival, allelopathic effects of *Centaurea maculosa*, Achillea biomass ratio with-competition to without-competition).

- **growth**: a measurement of the change in size (or a proxy of size) in a time period (where the starting point is not the birth, germination or equivalent), or a growth rate (irrespective of the starting point). It can be a relative or an absolute measurement of growth or growth rate. Irrespective of the environmental condition (pollution, stress, competition, herbivory/parasites, fertigated/non-fertigated, etc.), the measurements of growth or growth rate are classified as “growth”. If the trait is a change in growth (or another performance measurement) see “response”.

- **size**: Total body mass, or an overall aspect/measurement of body size as a whole (e.g. total length, plant height, thorax with, plant diameter, basal area, number of tillers (grasses)). Size of parts of the body (e.g. tarsus length, intracranial length, length of the longest leaf, etc.) are not included here but in “othermorphology”. In plants, measurements of the whole aboveground size (e.g. aboveground biomass, plant height, plant diameter or basal area, etc.), as well as measurements of the whole belowground size (i.e. root mass) are included as “size”. Other morphological features of roots (e.g. length, diameter, etc.) are classified as “othermorphology”. The “stem diameter” of tree species is a measurement of “size” but in herbaceous plants, it is “othermorphology”. “size” is not necessarily the final or average body size in a population, the measurements of size at a given age/stage (but not at birth/hatch or

equivalent) are also classified as “size” (e.g. Body mass lambs (kg)). Note that, similar situations can be classified differently according to the operational decision of the authors of the primary source of information, e.g.: in a common garden experiment for a short period of time if the authors report final biomass, the trait is classified as “size”, but if the authors report growth rate (biomass over time) it is classified as “growth”. If the trait is a change in size (or another performance measurement) see “response”.

- **othermorphology**: all remaining morphological traits. This includes the size of parts of the body (e.g. tarsus length, intercranial-length, etc.) or plant (e.g. leaf size). The size of a part of the body at a given age/stage is also “othermorphology”. It is independent of the environmental condition of the measurement, i.e. a measurement of tarsus length (or leaf size, etc) is always “othermorphology” independently of if it was measured in a controlled environment or in a polluted environment (or in another condition), and so on.

- **otherLH**: all remaining life history traits that are not classified as phenology or growth (e.g., offspring size, # of offspring, fecundity, survival, parental investment, reproductive effort/allocation, time to reach maturity or any stage in the life cycle). It is independent on the environmental condition of the measurement, i.e. a measurement of survival (or fecundity, etc.) is always “otherLH” independently of is was measured in a controlled environment or in a polluted environment (or in another condition), and so on. If the trait is a change in survival (or another performance measurement) see “response”.

- **response**: response to a number of abiotic and biotic factors. It is the change, relative change, or ratio of a measurement of performance (size, growth, survival, fecundity, etc) between two conditions (e.g. Size reduction—difference in size between pH 7 and 4, relative to size in pH 7—; Leaf Cpi—[no.leaves\_woc - no.leaves\_wc]/no.leaves\_woc—; Mycorrhizal response [inoculated\_biomass-Uninoculated\_biomass]/Uninoculated\_biomass).

## data\_type

The current categories are:

- **linear**: a measurement of length with a dimensionality of 1. A ratio between lengths (or a length over another variable) is not a “linear” variable because its dimensionality is not 1. A ratio between linear dimensions (e.g. length over width, or height over diameter), or a linear dimension over another variable (e.g. specific root length (SRL) [mm/mg]) is not “linear” because its dimensionality is not 1 (e.g.  $SRL = [1D]/[3D]$ , dimensionality -2).

- **area (2D)**: a measurement of area with a dimensionality of 2. A ratio between areas, or an area over another variable (e.g. specific leaf area (SLA) is not “area (2D)” because its dimensionality is not 2 (e.g.  $SLA = [2D]/[3D]$ , dimensionality -1).

- **cube (3D)**: a measurement of volume or mass or amount of matter with a dimensionality of 3. A ratio between volumes/mass, or a volume/mass over another variable (e.g. “Stem mass ratio (stem mass / whole plant mass)” or “Rhizome mass over length”) is not “cube (3D)”

because its dimensionality is not 3 (e.g. stem mass / whole plant mass= $[3D]/[3D]$  is a dimensionless proportion; Rhizome mass over length= $[3D]/[D]$ , dimensionality is 2 but it is not an “area (2D)” because it depends on the shape of the organ and the density of the tissue, so it is “other”).

- **count**: an amount of discrete units (e.g. number of leaves, clutch size, fecundity when number of offspring or propagules, etc.).

- **proportion**: the ratio between a part over the total (including that part). It can be expressed as a proportion (0-1), as a percentage (0-100). Some concentrations are also proportions, concentrations expressed as % M/M, % V/V, ppm, mg/g, ug/g (or similar). Molar concentration or  $\mu\text{mol/g}$  are not exactly proportions because the numerator and denominator are not in the same units but are still a measurement of the ratio between a part over the total and so, they were classified as “proportion”. A ratio between a given amount in a stressing/disturbing condition (e.g. biomass in a toxic environment or in presence of parasite/herbivore or competitors) over a control condition expected to express the maximum potential (e.g. biomass in a non-toxic environment or in absence of parasite/herbivore or competitors), was also considered a “proportion”. Similarly, the relative difference between these conditions was also classified as “proportion”, because it is a proportional change (e.g. tolerance to herbivory: (damaged plant biomass - undamaged plant biomass) / undamaged plant biomass). The difference is that in the first situation “transf\_data”=“raw” and in the last one “transf\_data”=“resid”. When the ratio is not a part over the total, this is not a “proportion” (e.g. root to shoot biomass, reproductive over somatic/vegetative biomass), this is a dimensionless ratio (i.e. “ad\_ratio”).

- **time**: the time needed for some process or event.

- **date**: a temporal landmark for a given event in a reference system with an arbitrary starting point (e.g. number of days from January 1st). A time in the day (e.g. hours from midnight) is the same type of trait.

- **temperature**: temperature in the scale and units reported in the primary source of information (e.g.  $^{\circ}\text{C}$ ,  $^{\circ}\text{K}$ ,  $^{\circ}\text{F}$ ).

- **rate**: a measurement of speed, an amount over time (e.g., growth rate, metabolic rate (cal/s), etc.). “rate” is a diverse category because the numerator can be “count” (e.g. eggs/day), “linear” (e.g. Growth (mm/day)), “proportion” (Grow(%/d)), “cube (3D)” (e.g. growth rate (pupal mass divided by preadult developmental period) OR Resting Metabolic Rate ( $\text{mg O}_2 \text{ h}^{-1}$ )), “ad\_ratio” (e.g.  $\text{RGR} = \ln[m_2/m_1]/\text{time}$ ), “other” ratios (e.g. Stomatal Conductance ( $\text{mmol m}^{-2} \text{ s}^{-1}$ )), etc. However, even when a trait is called “rate” in the primary source of information is not necessarily a rate according to the current classification (e.g. “pregnancy rate”, “survival rate”, “germination rate” are “proportion”; “Unit leaf rate ( $\ln [\text{mass\_t1} / \text{mass\_t0}]$ )” is an “ad\_ratio”; etc.).

- **ad\_ratio**: the ratio between two independent amounts with the same dimensionality is a dimensionless ratio. It is not a proportion because the numerator is not a component or fraction of the denominator (e.g. shoot:root ratio, C:N ratio, reproductive over vegetative biomass).

- **index**: ordinal variables and their derivations. However, when a trait is called “index” in the primary source of information it is not necessarily an index according to the current classification. For example, “index of ear size” $[(EL \times EW)^{0.5}]$  is a called index but it is quantitative variable proportional to the ear size, its dimensionality is 1. We classified this trait as “other”. Another example is “Gonadosomatic index (ova mass over body mass)” this is a “proportion” even when called index.

- **other**: other ratios and other quantitative variables not meeting the previous descriptions.

“data\_type” is particularly relevant when computing and comparing darwins ( $D$ ) among different data types. The rate of change darwins is the relative (to mean) rate of change of a given variable ( $X$ ) at a given time ( $dt$ ) (Haldane 1949). Darwins is:

$$(1) \quad D = \frac{d\left(\frac{dX}{X}\right)}{d(t)}$$

by integrating  $1/X$  on  $dX$  we obtain:

$$(2) \quad D = \frac{d(\ln(X))}{d(t)}$$

In a discrete interval of time ( $t$ ), we get:

$$(3) \quad D = \frac{\ln(x_2) - \ln(x_1)}{t}$$

The dimensionality of the trait affects darwins. As an example, if we have three traits total length ( $l$ ), body mass ( $m$ ), and mass of gonads over total body mass ( $p$ ). The relation between these three traits will depend on the allometric relationships, but it would be approximately as follows:

$$m = a \cdot l^3$$

$$p = \frac{b \cdot l^3}{a \cdot l^3} = \frac{b}{a}$$

So, the rate of change in darwins for these traits would be:

$$D(l) = \frac{\ln(l_2) - \ln(l_1)}{t}$$

$$\begin{aligned}
D(m) &= \frac{\ln(m_2) - \ln(m_1)}{t} \\
&= \frac{\ln(a \cdot l_2^3) - \ln(a \cdot l_1^3)}{t} \\
&= \frac{\ln(a) + \ln(l_2^3) - \ln(a) - \ln(l_1^3)}{t} \\
&= \frac{3 \cdot \ln(l_2) - 3 \cdot \ln(l_1)}{t} \\
&= \frac{3 \cdot (\ln(l_2) - \ln(l_1))}{t} \\
&= 3 \cdot D(l)
\end{aligned}$$

If the allometric relationships remain constant:

$$\begin{aligned}
D(p) &= \frac{\ln(p_2) - \ln(p_1)}{t} \\
&= \frac{\ln(\frac{b}{a}) - \ln(\frac{b}{a})}{t} \\
&= 0
\end{aligned}$$

As was shown in the previous example, the type of variable, by itself, strongly affects the estimation of evolutionary change in darwins.

#### transf\_data

If the trait measurement has been transformed and the means and standard deviations were computed on basis of the transformed data (e.g. body mass expressed as log(g) instead of grams, so the mean is the average of the logarithm of the body mass). The current categories are:

- **raw**: mean and sd computed from raw data (no transformation).
- **ord**: the trait is the score in the axis of an ordination analysis (i.e. PCA, CVA).
- **arcsin**: arc sin ( $\sin^{-1}$ ) transformation.
- **arcsin.sqr**: angular transformation, i.e.  $2 \cdot \sin^{-1}(\sqrt{x})$ .
- **resid**: residuals of regression or other statistical model ( $y - \hat{y}$ ), centered variables ( $y - \bar{y}$ ), absolute ( $Y_t - Y_c$ ) or relative difference ( $[Y_t - Y_c]/Y_c = Y_t/Y_c - 1$ ) between a treatment and a control.
- **ln**: mean and sd computed from natural log-transformed data. We also include data like  $\ln(x)/\ln(y)$  and  $\ln[x/y]/t$

- **log10**: mean and sd computed from log10-transformed data. We also include data like  $\log_{10}(x)/\log_{10}(y)$  and  $\log_{10}[x/y]$ .

#### **data\_scale**

- **ratio**: constant interval with a precise zero.
- **interval**: constant interval with an arbitrary zero (e.g. temperature, time of day, rescaled data: Principal components or discriminate functions).
- NA (**empty cells**): undefined scales and ordinal scales (i.e. “data\_type”=“index”) because ordinal scales do not have a constant interval.

Darwins cannot use interval data, but Haldanes can (because it’s rescaled units, essentially removing an absolute zero point). Ratio data are actually interval data, but with a clearly defined zero.

#### **n1**

Sample size for sample 1.

#### **n2**

Sample size for sample 2.

#### **mean1**

Mean of the phenotypic trait for a given system for sample 1.

#### **mean2**

Mean of the phenotypic trait for a given system for sample 2.

#### **sd1**

The standard deviation around the mean of the phenotypic trait for sample 1. Can be calculated manually if necessary.

#### **sd2**

The standard deviation around the mean of the phenotypic trait for sample 2. Can be calculated manually if necessary.

#### **out\_in**

Population(s) outside or inside its native range. Of your two data points (samples) to calculate the rates – if one of the datapoints is outside the native range, it is considered “out”. Natural range expansions are considered “in”. You can have an introduction within the native range which would be “in”. Invasive species, with range expansions, would be “out”.

**urban\_disturbance**

Category used in Alberti et al. 2017 PNAS that classified urban disturbance category (decision tree in Figure S2):

- **Biotic**: “biotic interactions stem from introductions, and are subcategorized depending on the study organism's ecological role: introduced species, or species in its native range responding to an introduction. Introduced species were further subdivided into instances where the introduced species adapts immediately following introduction, vs adaptation after range expansion (post introduction)”.
- **Habitat Mod**: “changes due to climate change in general, modification of the landscape, or pollution”.
- **Heterogeneity**: “heterogeneity may refer to micro-habitat or micro-climate, and can refer to heterogeneity in space or time”.
- **Novel**: “Novel disturbances were defined as those disturbances to which organisms respond to with novel adaptations, such as the rapid evolution of zinc tolerance”.
- **Social**: “Social interactions refers to disturbances with are direct and intentional result of human agency”.

**disturbance**

It is a classification of the environmental change (or driver/disturbance) driving the phenotypic change, based on the categories used by Hendry et al. (2008) and Palkovacs et al. (2012). The current categories are:

- **Climate change**: this disturbance can often be difficult to clearly identify. A disturbance is classified as “Climate change” when the primary source of information explicitly considers climate change as a potential effect, or when the driver of the system is a climatic event (e.g. *Brassica rapa* (field mustard) changed flowering time after drought in Southern California, USA). Experimental systems manipulating CO2 levels were considered “Pollution” (See “Pollution” definition).
- **Hunt\_harv**: any case of hunting or harvesting of a species by humans.
- **Introduction**: “when humans transferred a species to a new geographical location, and comparisons were then made between introduced and ancestral populations”. Also, range expansion after introduction (i.e. “when humans introduced a species to a new geographical location, and the species then spread on its own accord to occupy multiple sites. Comparisons were here made among the self-colonized populations”).
- **Pollution**: an environmental change is classified as “Pollution” when chemical substances (e.g. heavy metals, pesticides, etc) are added or released into the environment contaminating, dirtying, and/or making harmful an area (usually air, water, or soil) where the focal species occur. The environmental change resulting from changing the natural concentration of

substances present in the environment (e.g. a change in the level of CO<sub>2</sub> in the air or a change in the level of nutrients in the soil) or a physical parameter (e.g. temperature, pH, etc) is also categorized as “Pollution” even if not harmful to the species being studied.

- **Landscape change:** It is a habitat modification (different from “Pollution” and the introduction of a new species) in the patches occupied by the population(s) being studied. It can be the creation/restoration of new patches/habitats for the population(s) (e.g. a newly formed freshwater pond for an aquatic species, newly restored habitat), changes in the connectivity among patches/habitats (e.g. the construction of a dam in a river or the construction of a fish ladder, level of fragmentation of the habitat), or a change in the land-use/quality of the patches (e.g. forest clearing, livestock/herbivores inclusion/exclusion, newly irrigation system).

- **Response to introductions:** when a species (native or alien) is responding to the introduction of a new species. The newly introduced species can be a competitor, a predator, a parasite, a new host or prey, etc. The comparison is established among populations (or times, according to the design) of the responding species (not the newly introduced species). When the responding species is native, “out\_in”=“In”; when it is alien, “out\_in”=“Out”.

- **Other:** It is “in situ natural variation” and “Self induced range or host expansion”, i.e. when native populations were not subject to an obvious human impact, most studies in this category involve the long-term monitoring of natural populations, such as Darwin’s finches of the Galápagos (Grant & Grant 2002, 2006); also, when new populations were established without any direct human influence, and comparisons were then made among the new populations or among the new and ancestral populations (Examples include birds colonizing new habitats within their native range—Yeh & Price 2004—or new islands outside their native range—Clegg et al. 2002—).

#### env\_change

- **ongoing:** ongoing environmental change. The environmental change has been occurring through time and the phenotypic change was measured in a temporal window within this process (but not from the starting point).

- **novel:** novel environmental change. The environmental change has a defined starting point and phenotypic change was measured from this point.

This category applies only to allochronic studies, synchronic are always “env\_change”=“novel” (because the proposed ancestral state is placed previous to the drivers happening). When “disturbance”=“Other”, if the driver it is unknown, “env\_change”=“NA” (empty cell). But in some cases the driver of change is known, like “self induced range expansion”, also experimental systems can represent “in situ natural variation” but from a known starting point. In these cases, the value of “env\_change” can be chosen.

### **Introductions**

This is an arbitrary way of ranking systems with more than one disturbance/driver. It ranks from 0 to 5, with 0 being the disturbance not pertaining to the system at ALL, and 5 being very important to the system. The current method for ranking is:

- 1 - Range expansions without invasions
- 2 - Range expansions with invasions (but invasions not specified)
- 3 - paper specifically investigates response to an invader or introduced species
- 5 - paper specifically investigates an invader or introduced species

### **Other**

This is an arbitrary way of ranking systems with more than one disturbance/driver. It ranks from 0 to 5, with 0 being the disturbance not pertaining to the system at ALL, and 5 being very important to the system. The current method for ranking is:

- 1 - Paper specifically investigates climate change
- 5 - Long term monitoring, even for commercially exploited species (e.g. salmonids), but not specifying hunting/harvesting
- 5 - Paper was in situ natural variation

### **resp\_introduced**

This is an arbitrary way of ranking systems with more than one disturbance/driver. It ranks from 0 to 5, with 0 being the disturbance not pertaining to the system at ALL, and 5 being very important to the system. The current method for ranking is:

- 1 - authors note that invasions happened and could be a source of change, but do not explicitly consider it (do not add number to invasions, but do add to landscape change)
- 4.1 - response to removal of invaders
- 5 - response to introduced organisms

### **Hunting\_Harvesting**

This is an arbitrary way of ranking systems with more than one disturbance/driver. It ranks from 0 to 5, with 0 being the disturbance not pertaining to the system at ALL, and 5 being very important to the system. The current method for ranking is:

- 1 - Looking at herbivory in plants
- 1 - predation on organisms was specifically considered, but natural predation

- 2 - predation on organisms was specifically considered, but introduced predation
- 2.1 predation on organisms specifically considered but REMOVAL of predators due to natural causes
- 3.1 predation on organisms specifically considered but REMOVAL of predators due to human influences
- 4 - Species is a commercially or recreationally exploited species (e.g. salmonids), but the study is not specifically considering hunting/harvesting
- 5 - Paper looks directly at targeted hunting or harvesting by humans

#### **landscape\_change**

This is an arbitrary way of ranking systems with more than one disturbance/driver. It ranks from 0 to 5, with 0 being the disturbance not pertaining to the system at ALL, and 5 being very important to the system. The current method for ranking is:

- 1 - Paper investigates range expansions w/o invasions
- 2 - Paper investigates invasions, but not range expansion
- 2.1 - paper investigates removal of organisms, but not range expansion
- 3 - Paper investigates range expansions w/ invasions (this includes range expansion into more lakes (for fish) via introduction)
- 4 - Paper investigates pollution/biocontrol/urbanization
- 5 - Paper specifically investigates landscape change

#### **Pollution**

This is an arbitrary way of ranking systems with more than one disturbance/driver. It ranks from 0 to 5, with 0 being the disturbance not pertaining to the system at ALL, and 5 being very important to the system. The current method for ranking is:

- 1 - pollution is considered as potential cause of observed phenotypic changes
- 2 - historical fluxes in a pollutant are mentioned in article (e. g. copper flux due to historical copper mining, eutrophication)
- 5 - Paper specifically investigates pollution/biocontrol

#### **climate\_change**

This is an arbitrary way of ranking systems with more than one disturbance/driver. It ranks from 0 to 5, with 0 being the disturbance not pertaining to the system at ALL, and 5 being very important to the system. The current method for ranking is:

- 1 - Paper was in situ natural variation
- 2 - Paper does not mention climate change but talks about environmental variation that can be attributed to climate change (e.g. fluctuation in annual precipitation)
- 2.1 - time series for commercially exploited species (e.g. salmonids)
- 3 - Paper specifically investigates range expansion (but not invasions specifically, but if invaders expanded, goes here)
- 4 - Paper looks at thermal tolerance (natural or induced, e.g. warm water effluent from power plants)
- 5 - Paper specifically investigates climate change

**time\_series**

Point out if this case is related to a time series that can be included in the TimeSeriesDB of PROCEED.

**tserieshandling**

It states how the time series was handled to take two samples to add to the RatesDB.

**comments**

Specify the source of information for each rate (e.g. data source within the paper: figure 3a). Other notes that might be useful to someone looking at the database.

**DOI**

Digital identifier number.

**reference**

This is the reference of the primary source of information, which is given in the following format (all one line): Last name of the first author and year, Journal (standard abbreviations) Volume: Page numbers or article identifier (e.g. Cox 1997, Can. J. Fish. Aquat. Sci. 54: 1159-1165).

**reference\_comments**

Comments related to the primary source of information.

**copyright**

If the authors of the manuscript do not want us to share their data or any other information with regards to copyright.

**who**

Who has entered the data (your name).

**pending\_issues**

Any unresolved issues that need to be addressed.

**v5\_taxa\_fine**

This is an old variable kept from previous versions of the database. For future versions, there is no need to code new data entries according to this variable. Currently, this variable has been replaced by “class” and “order” for more precision.

**v5\_Roff\_Mousseau**

This is an old variable kept from previous versions of the database. For future versions, there is no need to code new data entries according to this variable. The current version of this variable is “trait\_type”. The last definition of “Roff\_Mousseau” was:

*Class of traits as determined by Roff and Mosseau 1987.*

- Morphological: e.g. body size, wing size, other metric characters.
- Behavioural: e.g. alarm reaction, activity level, sensitive to conditioning.
- Physiological: e.g. oxygen consumption, resistance to heat stress, body temperature.
- Life-history: Fecundity, phenology, viability, survival, development rate, “directly related to fitness”

**v5\_Kingsolver\_Diamond**

This is an old variable kept from previous versions of the database. For future versions, there is no need to code new data entries according to this variable. The current version of this variable is “trait\_type”. The last definition of “Kingsolver\_Diamond” was:

*“Class of traits as determined by Kingsolver and Diamond 2011. Current classifications are:*

- size: Overall aspect of body size (body size, length, mass, PCs for body mass). Size proxies are NOT included here (e.g. tarsus length, intercranial-length, etc.)
- othermorphology: all remaining morphological traits. This includes morphological components that are used as proxies for size (e.g. tarsus length, intercranial-length, etc.)
- phenology: the timing of life-history events (e.g., photoperiod, initial date, peak date, or duration of events such as germination, flowering, laying date, and hatching date)
- otherLH: all remaining life history traits (growth rate, offspring size, # of offspring, survival), if biotic interactions (e.g. competition), then classified as otherLH, if abiotic stressor,

*classify as physio. For herbivory, see physio – will depend on if it is resistance or tolerance to herbivory.*

*- behaviour*

*- physio: metabolism, bite force, growth in stress (abiotic factor), resistance to herbivory (e.g. a chemical defense; tolerance will be based on trait), diet: trophic level, diet, food type, if both abiotic and biotic interactions, classify as physio.”*

#### **v5\_data\_type**

This is an old variable kept from previous versions of the database. For future versions, there is no need to code new data entries according to this variable. The current version of this variable is “data\_type”.

#### **v5\_driver**

This is an old variable kept from previous versions of the database. For future versions, there is no need to code new data entries according to this variable. The current version of this variable is “disturbance”.

#### **v5\_disturbance**

This is an old variable kept from previous versions of the database. For future versions, there is no need to code new data entries according to this variable. The current version of this variable is “disturbance”.

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